

GeoZome

Mapping the Geopolitical Rhizome

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INTRODUCTION

Concept

What I am moving towards is the creation of a geopolitical mapping tool. A tool to assist in the *Reverse Orchestration*¹ of geopolitical conflicts and events through the synthesis of temporal, spatial and social mapping. Encoded within this mapping tool an *ontology*², epistemology³ and methodology⁴ for compiling, plotting and notating geopolitical choreographies.

Through the mapping, I aim to extract novel information than that originally inputted. Employing graphic techniques to make structures, links and rhythms *emerge*⁵ through the visual rendering of more local and fundamental agents and actions.

To make a bird and observe a flock.

¹ **Reverse Orchestration** - The forensic practice of reconstructing (through Research and Mapping) geopolitical events in history.

² **Ontology** - The study of what exists; the search for categories we can sort existing things in.

³ **Epistemology** - The study of how it is we gain knowledge. The systems, processes and actions involved in the knowledge digestion.

⁴ **Methodology** - The procedure followed in order to understand geopolitical events.

⁵ **Emergence** - The quality of complex behaviours arising from local and simple interactions.

Geopolitical events occur in spaces, across times and between varying entities. In the contemporary mode of communicating these events, we are exposed to merely text fragments and single and/or limited frames. We are presented with meticulously manicured interfaces—abstractions and manifolds— of these spaces, times and entities. Through the compression this communication requires, any continuance and permanence of these elements is prohibited. They are made ephemeral and spectral, materialising only momentarily (if convenient and if the bounds allow for it), then phasing out of existence.

My research aims to answer the following question:

How can visual mapping software foster a deeper understanding of historical geopolitical events?

My research will also attempt to justify why I felt compelled to seek its answer.

It will do so using a triptych format. Wherein:

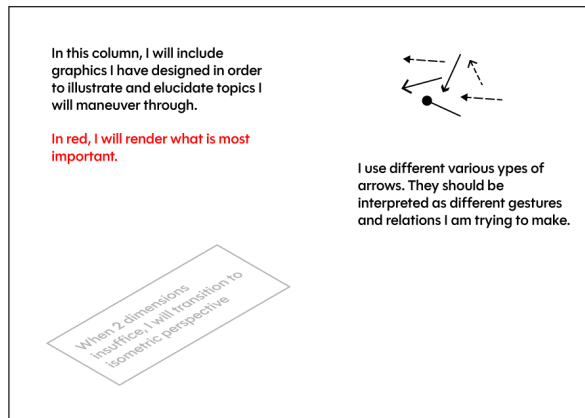


Figure 0 - Defining what the left-most column will contain.

This column will act as its spine—grounding the writing in the research I conducted in theory relevant to its scope.

To answer this primary question, I will first provide a response to its constituent sub-questions. Each will be answered in the form of a postulate.

- *How do these domains (social, temporal and social) differ in their representations of position? What would “mapping” these positions look like?*
- *What data structures and frameworks are fit and apt for representing geopolitical relationships and dynamics?*

In this column I will use the example of the Trump administration’s shutdown of USAID—at the time the global largest foreign aid agency. This research is not *about* the USAID shutdown but I will use it as a contemporary instance of a geopolitical event. One that we can dissect and see how it can be mapped.

In **Postulate I**, I will explore the qualities and models we can use to define and conceptualize these positions. Unified, they compose what I will refer to as “*Position*”.

In **Postulate II**, I will argue that these *Positions* collate to form a geopolitical rhizome, what I will refer to as a “*geozome*”.

→ *How can geopolitical research retain its complexity and nuance, whilst adhering to a model intelligible to both the user and the software? How does currently available mapping software struggle with this?*

Methodology

The methodology followed to develop GeoZome took many forms and traversed many fields and perspectives. From a Graphic Design perspective, I sought relevant work from experts in the fields of Data Visualization, Cartography and Symbolology. These fields definitively intersect with Geospatial studies and Political Sciences. This provided a way to situate GeoZome within existing practices thus, be imparted the rigors and standard factored within their respective fields. Through the revision of relevant literature and direct consultation with experts for feedback and insights, I was able to ground my research and work to appropriate domains.

In **Postulate III**, I will posit that to understand a geopolitical event is to navigate its geozome. I will introduce my mapping software: “GeoZome” as a navigational interface for the geozome.

POSTULATE I

That geopolitical information has a *Position*

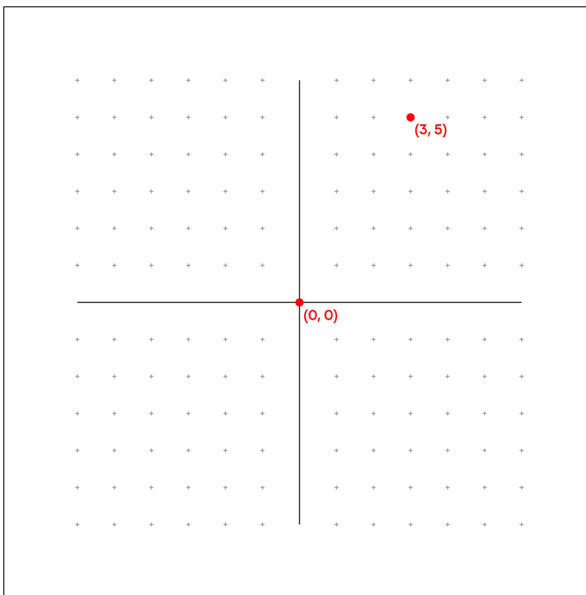


Fig. 1 - Plotting of coordinate (3,5) and origin (0,0)

Spatial Position

Position is a relative concept. To have a set of coordinates is to imply an origin to give meaning to those coordinates. (3, 5) means very little until we agree that (0,0) is our common origin.

The common understanding of position tends to be affixed to cartographic and geometric understandings. This is limiting and renders our scope too narrow to accommodate the information contained within geopolitical events. For within this understanding, we are bound by the need for a universal origin. Implied within this origin, are the axes and the grid of

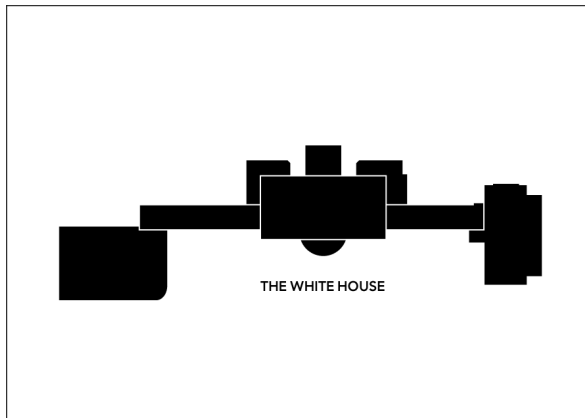


Fig. 2 - The White House

the underlying space on which to plot. This works well with information that is easily quantized.

Coordinates are not the only method to represent spatial positions. In fact, these bi-numeric coordinates are oftentimes too specific for particular use cases. Approaching these positions with the concept of scale in mind reveals these shortcomings.

To demonstrate the constraint of viewing position as singularly a spatial term, I will attempt to locate the American White House. The White House is a piece of infrastructure, easily located spatially using latitude and longitude coordinates. This position inhabits a pre-existing space; one with agreed upon axes and origin.

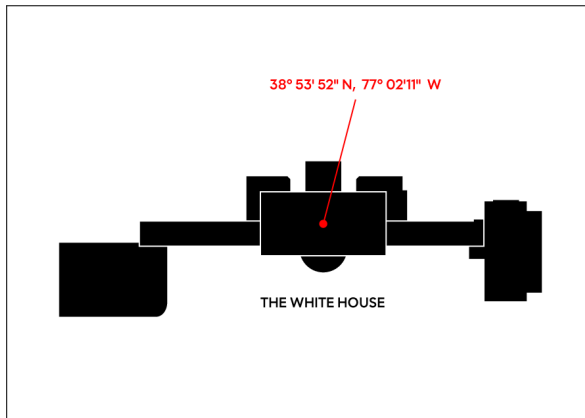


Fig. 3 - The White House with labeled coordinates

Let's say I am to give a single coordinate to represent the location of the White House. For lack of a better method, I can take the center of the building as its coordinate: $38^{\circ}53'52''N$ $77^{\circ}02'11''W$. As the White House is not a singularity, it can be subdivided into its constituent subparts.

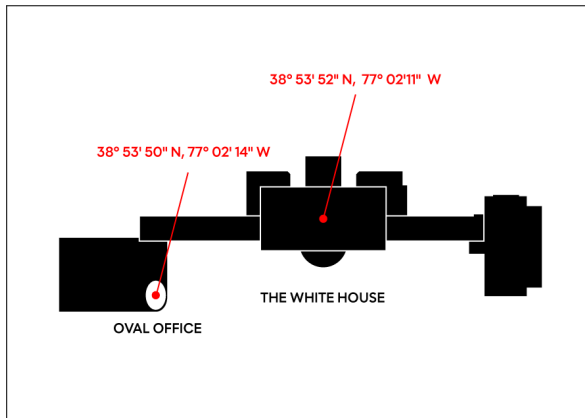


Fig. 4 - The Oval Office labeled with coordinates

This vessel contains smaller vessels within it.

The Oval Office is a room within the White House. Using our coordinate representation I have no method of communicating their relationship (that one is located *within* the other) or differences in scale without additional coordinates. Given additional coordinates I could then demarcate the bounds of the White house, as to provide a container in which I can insert the Oval Office. But if I can zoom in, from the White House into the Oval Office, I can too zoom out.

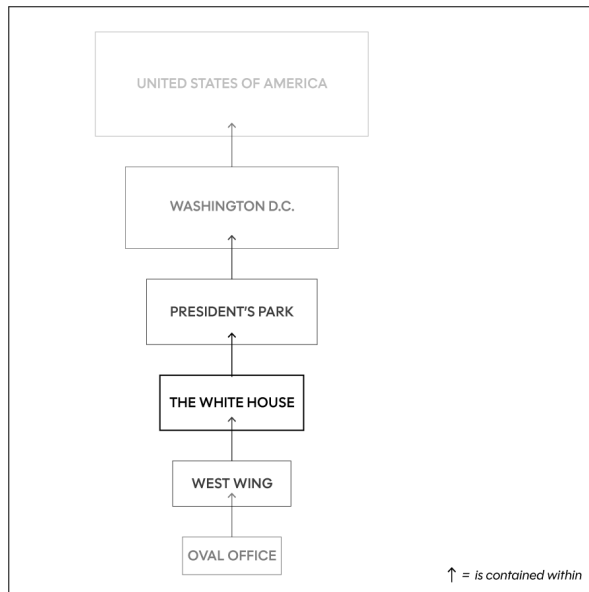


Fig. 5 - Hierarchical identity-model spatial position

This vessel is contained within a larger vessel.

As the White House is not a totality, it is itself, a subdivision of some larger whole. The white house is located within President's Park which is itself located within Washington D.C., which is located in the USA etc...

To represent these *parthood*⁶ relations I required a system which can both merge and bifurcate; a hierarchical model of sorts. A model whose positions are dictated not via

⁶ **Parthood** - A term in the field of Mereology describing the quality of being a component-part of a larger whole. A hand has finger-parts and thus a finger shares a parthood relation with the hand.

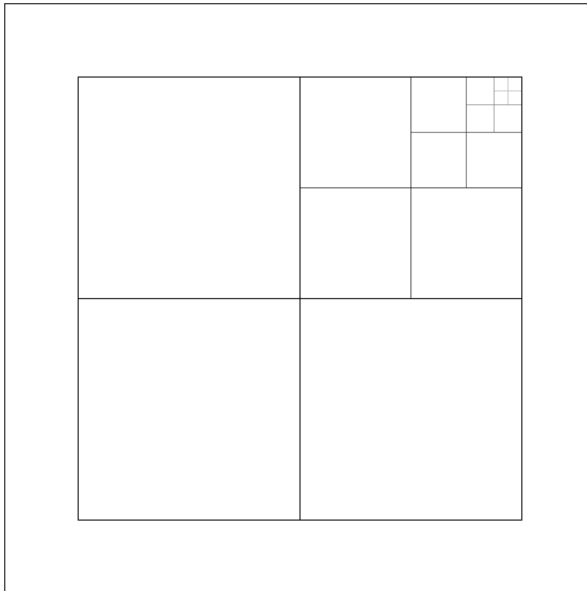


Fig. 6 - Quadtree

transformation but via segmentation and dissection. Such models exist and find vast application throughout computer science.

Take the Quadtree as an example of a hierarchical model for spatial positions. The underlying premise behind the quadtree is fractal: that a square can be divided into quadrants, and that a quadrant can be further divided into quadrants itself and so on, ad nauseam. Any position within a quadtree thus defined by the sequence of quadrants leading up to it. In some sense, I have sacrificed *specificity* for *context*. In so, viewing positions not merely as vectors of ***displacement***, but also as vectors of ***identity***. Although both describe the same piece of information, each positional model offers differing affordances. With the displacement model, I can ascertain distances and areas but will be unable to gauge how these various positions are related without the additional employment of an identity model for position.

I am now left with 2 spatial locations for the White House. One derived from displacement from an origin (our coordinates). The other,

Temporal Position

The spatial position is of second nature to us. Up, down, left, right. However the universe is not only space, but also time: spacetime. This second thread, from which the fabric we exist within is woven (time), offers another position I can stack on top of the first. I now have a *Position*⁷. A voracious term that seeks to encompass all the various positions on all the various dimensions contained within a single piece of information.

derived from identity and parthood (our tree). I can now locate the White House in terms of 'where' but also 'within what'. Combining these two systems allows for a more tangible method for relaying territorialization—a major concept within geopolitics.

With this transition from position to *Position*, I must now accordingly update what I mean by 'locating' the White House. I know I am now not only trying to ascertain its spatial position, but where it exists in time.

⁷ **Position** – A term which compounds the various attributes a piece of information can be quantized into across various metrics. In such, each *Position* contains within itself, various positions of different kinds.

In some ways, the temporal can quite easily be thought of in terms of positions once I conjure the primitive concept of a numberline. Time can be represented as a 1-dimensional line extending from the beginning of time until now. This line must be terminated once the present is reached as projections into the future i was not within our current grasp at the moment. As time and space are not only related, but essentially the same concept, everything that has existed *somewhere* in space must have also existed *somewhen* in time.

The physical axiom of the conservation of mass tells us that matter can be neither created nor destroyed. However, plotting everything as having always existed is rather disadvantageous for our purpose of temporal mapping.

Everything has always existed, but not in the same *arrangement*. It is the act of having the components in a defined order that incites its inception. A 'pancake' is not a

The atoms that make up the cement that went into the construction of the White House have always existed. The Big Bang can not be the point of inception for everything. What I need is a formalized standard for how the temporal points of entry and exit will be dictated.

'pancake' until the ingredients meet the *Order Requirement*⁸ and *State Requirement*⁹. To arrive at a 'pancake,' its ingredients need to be first mixed together (*Order Requirement*) and subsequently cooked (*State Requirement*).

Returning back to the White House, I have an option between two dates I can deem its point of temporal inception.

- A) Construction started October 13, 1792
- B) Construction finished November 1, 1800

Given our two requirements of Order and State, I can eliminate option A as neither conditions are met. On October 13th 1792, the component-parts of the White House were neither Ordered (in their final positions) nor did they possess the right State (able to offer shelter). This leaves us with November 1, 1800 as the White House's date of inception. Given the White House continues to exist, I am unable to give the White House a date of 'termination'.

Given these metrics for deciding the appearance of certain entities in the temporal map, I can then derive how their

⁸ **Order Requirement** - The manner in which component-parts are organized to create a whole.

⁹ **State Requirement** - The attributes required of component-parts to constitute a whole.

disappearance is decided. However, greater 'momentum' is required to remove than is required to create. That is to say, that while our conditions for inception require both Order and State requirements to be met simultaneously, conditions for their removal require both requirements to be simultaneously broken.

Social Position

Having added both the spatial and temporal components to our *Position* for the White House, I can now attempt to find its social component.

Let me first clarify what I mean by a social component, by compiling what was needed to form its spatial or its temporal counterparts. We communicate these positions in terms of agreed-upon 'units'. The "minute" is a temporal unit, and the

A single window can be removed from the White House and it still remains the White House. It is once the Order of its parts are altered beyond a threshold— one disruptive to their State—that the White House ceases to be the White House.

"meter" is a spatial unit. This begs the question: *What are the units for this social component?*

In a sense, the search for the social component poses a novel obstacle whose absence was taken for granted in both the spatial and temporal positions. This is due to the lack of an agreed upon set of 'units' we can communicate in regarding the social position. Attempts at solidifying this 'unit' system have, however, been made.

I will introduce a model for such a system through recounting it's contribution to the monumental feat of the *Human Genome Project* (HGP). The HGP was a project designed for the mapping of all the genes in human DNA in order to ascertain their role in the body. The human genome consists of around 3 billion nucleotide pairs (National Research Council). A staggering amount of information considering the project began in 1990. To process and make meaning out of all

To give the White House a single 'unit' for what type of thing it is, will flatten all its other functions. Looking specifically at the White House:

Is the White House the building itself?

The location it is in?

People both live and work in it, so is it a residential building or a governmental one?

If the current one is nudged to the left a bit and a new one is built where the old one stood, which one, then, is the White House?

of this information, they had to provide it structure. This structure evolved but by 2004, the year the HGP was completed, it took the form akin to what is now referred to as a *Semantic Triple*¹⁰.

¹⁰ **Semantic Triple** - A model-or rather-grammar for contextualizing data; consisting of 3 parts:

Subject $\rightarrow \{Predicate\} \rightarrow$ Object

i.e.

i.e: France $\rightarrow \{Has Capital\} \rightarrow$ Paris

Hand $\rightarrow \{Has Finger Count\} \rightarrow 5$

Red Blood Cell $\rightarrow \{Has Function\} \rightarrow$ Transport Oxygen

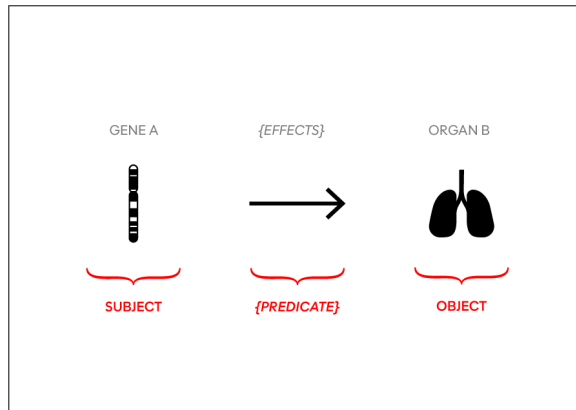


Fig. 7 - Example of a Semantic Triple

The Semantic Triple grammar granted me a formalized method in which I can begin to make connections between different pieces of information. Geopolitically, this could range from involved persons, infrastructure, resources etc..

For example, one could construct the following semantic triples about the White House:

White House → {is color} → *White*

White House → {contains room} → *Oval Office*

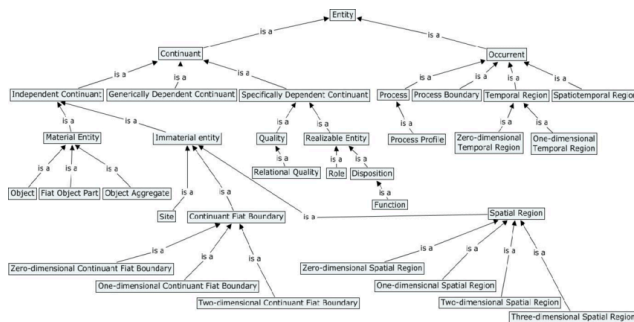


Fig. 8 - Basic Formal Ontology (BFO)

Barry Smith, a polymath known for his work in mathematics, philosophy and most notably *ontology*, joined the Human Genome Project to assist them in structuring the vast amount of data collected. Smith's stint at the HGP pivoted their approach from just solely an attributive semantic triple, to a formalized ontology—an early predecessor to the Smith's 2015 *Basic Formal Ontology* (BFO).



Fig. 9 – “One and Three Chairs” (Kosuth)

The BFO provides a model for classifying any *entity*¹¹. Wherein an entity can represent any real thing. Each of Kosuth’s 3 chairs is an entity. The BFO thus, provides us a roadmap I can follow to allocate any entity to one of the BFO’s *terminal classes*¹². It takes the form of a root system consisting of nodes and edges connecting these nodes. One begins at the top and works its way down the root.

Using the BFO, I could try to ascertain where the White House lies within its root-system. As everything is an entity, I will begin there and work my way downwards. To decide on the next node, I will consult the definition of its children nodes to see which to descend down to as outlined in the BFO’s specification guide. Each ontological category forms a semantic triple with its parent category. Such that:

[Child] → *is a* → [Parent]

¹¹ **Entity** – Any real thing. People, news articles, elections, Earth are all types of entities.

¹² **Terminal Class** – An ontological category with no child categories.

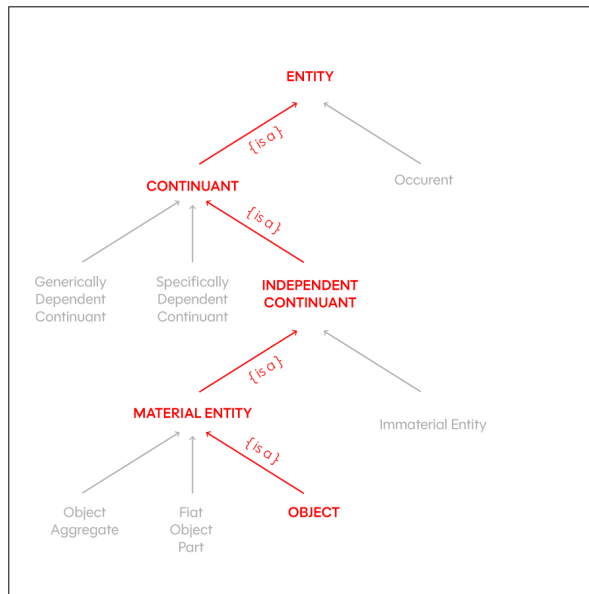


Fig. 10 - Locating the White House's position on the BFO

There is an indispensable advantage of approaching the various entities within geopolitics through an ontologic lens. That is, it allows for the information to be both *formalized* for the sake of the program and the usability, and *interoperable* between users, allowing for the dissemination and sharing of research. In addition to the glaring benefit from a software development viewpoint, it directly incites the epistemological intent of this program. That to understand a geopolitical event a contextual but nevertheless consistent framework is rudimentary.

[Entity] → [Continuant]

The White House falls under [Continuant] because it can be divided into spatial regions (Smith, 17).

[Continuant] → [Independent Continuant]

I can then posit it is an [Independent Continuant] as it persists within the temporal bound it exists in whilst not depending on any other [Entity] (Smith, 27).

Notice how by classifying the White House as an Independent continuant, I have reaffirmed I am talking about the building itself. That is not to say I have discarded its governmental functions. They will be reintroduced later. For now, I am talking about its husk.

[Independent Continuant] → [Material Entity]

As it is composed of "some portion of matter" (Smith, 28), I can categorise it as a [Material Entity].

[Material Entity] → [Object]

It is imperative that I clarify my chosen magnitude of scale. If presented with a satellite image, I could posit it as a singular [Object]. However, if instead I am given an

image of the Oval Office, I could dismember the White House into its constituent parts, rendering it an [Object Aggregate] (Smith, 30).

The cascade of categorization must terminate somewhere for my own sake. This process could go on until I have splayed all the quarks comprising the White House as individual [Objects].

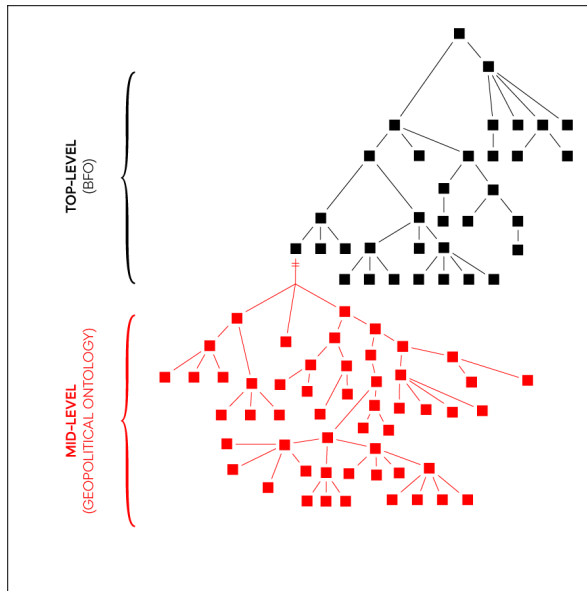


Fig. 11 - Appending the BFO from its [Object] Universal with a supplemental geopolitical ontology.

The BFO provides only the most general of categories. Smith often likens it to a power outlet. The BFO is considered a “Top-Level Ontology” as it implies and anticipates expansion. As our focus lies in Geopolitics, I will need to append the BFO. This became clear once I began to consider the generality of the [Object] class. I must differentiate between [Objects] that have distinct functions.

I can ‘plug in’ my geopolitical ontology at various points on the BFO’s existing root structure. Thus, providing a Mid-Level Ontology situated beneath the BFO. During a conversation I had with Data-Journalist Gerlinde Schuller—regarding my ontological approach—the existence of possible unidentified use cases was broached (personal communication, 16 May 2025). After our discussion, we established that facilitating the inclusion of user-built Bottom-Level Ontologies offers an opportunity for future development.

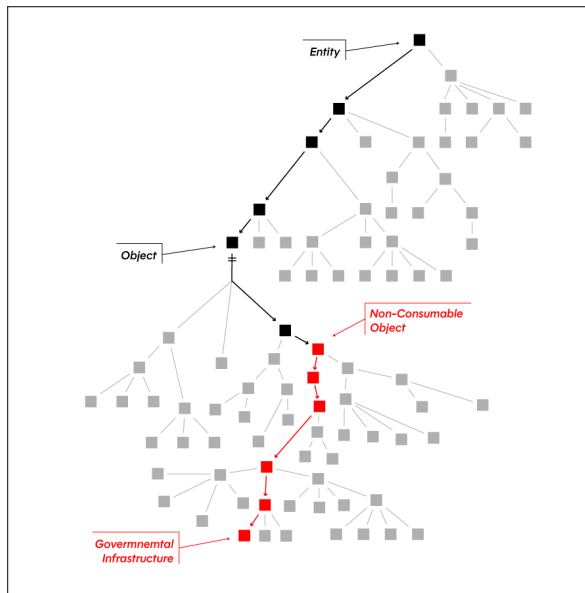


Fig. 12 - Parent to child inheritance of attributes.

Every terminal class is then defined by the chain link of parent classes it pertains to. In so, meeting the specifications of all these classes. Thus, any properties and functions of its parent classes are bestowed upon it.

5 parent classes up from the “Political Infrastructure” Universal¹³ (of which the White House is a Particular¹⁴ of) we can find the [Non-Consumable] Universal. As the name suggests, it is defined as entities which are not consumed in use. Its properties are syllogistically transferred to its children, allowing the assertion:

“The White House is not consumable.”

The extent in which supplemental classes will be appended to the BFO can not exceed the threshold between Universals and Particulars. The *kinds/categories* of objects there are are Universals, whereas the *unique instances* of a kind of object is a Particular.

¹³ **Universals** – Denoted via encasement in brackets (i.e [Material Entity]) The general ‘types’ entities can be. [Planet] is a universal; ‘Earth’ is not as it is a unique instance of the [Planet] type. Also referred to as Classes

¹⁴ **Particulars** – Unique instances of a universal. The toothbrush in your home is a specific instance of the universal [Toothbrush]. Also referred to as *Instances*

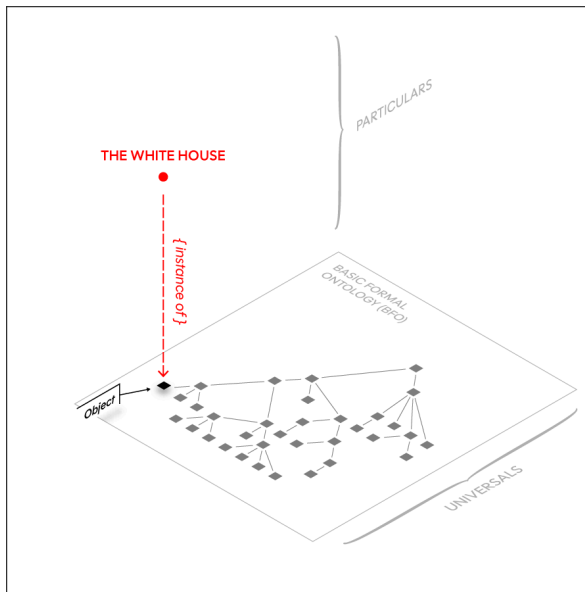


Fig. 13 - The White House is an instance of [Object] Universal in BFO

Employing an ontology, I have gained a rudimentary vocabulary and grammar for describing not only *what* things are, but how I can form connections between them. The task is then to conduct an analytic census—itemizing what Universals are required to grasp the underworkings of geopolitical events.

Within the BFO, the White House is a *Particular* of the [Object] *Universal*. It is a singular instance of a kind of Object. It can be useful to imagine the BFO as a reference or catalogue—comprising all *Universals*. Every Particular, thus, necessitates a *Universal* of which to be an instance of. The *Semantic Triple* grammar can now be used to denote what the White House is instantiated from in the following manner:

The White House $\rightarrow$ {instance of} $\rightarrow$ [Object]

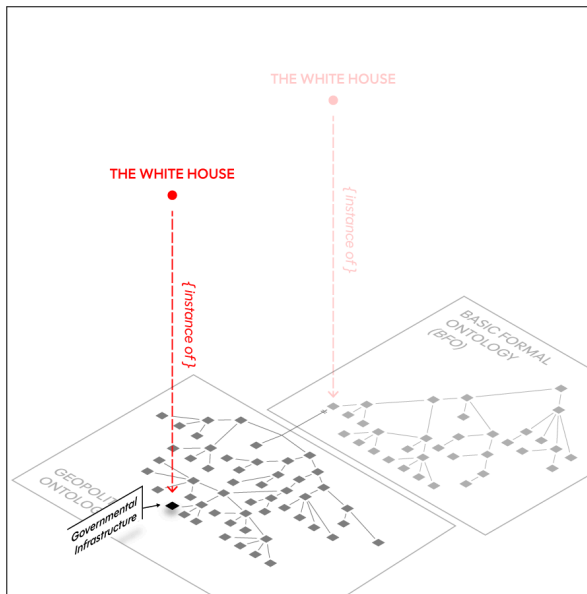


Fig. 14 - The White House is an instance of the [Governmental Infrastructure] Universal in my geopolitical ontology.

We lack a “dictionary” equally as rigid, rigorous and codified as those available for time and space—like the Gregorian calendar or the WGS84—in asking *what?* as compared to *when?* or *where?* Instead, I am compelled to construct my own. Having constructed it I can locate the social component of any geopolitical entity’s *Position*. A social position is not *quantized*, but rather *qualified*; qualified through the ontological “units” of their instantiation of Universals and connections to other Particulars.

As I have appended the BFO outwardly from the Object class, it is now necessary I provide the White House with an even more precise placement on our geopolitical ontology. Following the definitions at each intersection, the White House can be then classified under [Governmental Infrastructure]. Resulting in our updated Triple:

The
White House $\rightarrow \{instance\ of\} \rightarrow$ [Governmental
Infrastructure]

POSTULATE II

That these *Positions* construct a *Geopolitical Rhizome*

Geopolitical Rhizome

It is paramount I tread consciously about universalizations. Any frame in anthropocentric history warns us of the dangers of *classifications*. Forewarning us of the potential peril imposed on political Universalizations, Sergei Prozorov does not hesitate to brandish the phrasing "*violent project*," within the preface of his work: *Ontology and World Politics*. Prozorov is troubled by the *topology* of the Universal dimension. It is planar, reductive and insufficient—unfit through its compression of any particularities. The process of Universalization is to, as Deuze and Guattari articulated, "*—fabricate a beneficent God to explain geological movements*," (2).

To avoid such hazards, I must depart from the arboreal and hegemonic structure of the Universals. I must move towards a model that when adhered to elucidates nuance rather than disposes of it. Deleuze and Guattari provide such a model—one that is continually undergoing the process of construction rather than its construction occurring primordially. They entitle this model the *Rhizome*¹⁵: an alternate formation of a root system that does not branch downwards from a single point. In contradistinction, the Rhizome's root-system operates *laterally*.

It is important to state that the Rhizome is a living construct. It has tendrils which constrict, contract and seek. They seek to expand not out of voracity, but for the same reason a drop of ink disperses and disseminates in water—*entropy*. It has no center because there simply isn't one. I can momentarily manufacture one by picking an object and collapsing the position of everything else. In so, rendering them to appear as spokes, stemming out radially.

¹⁵ **Rhizome** – A term borrowed from Deleuze and Guattari, it describes a model of thought that emphasizes connections and relations over absolute positions or proximities. Its constituents are connected through lines of “varying slowness and viscosity” (2). These lines (semiotic chains) are biunivocal—*viral*—in that every arrow pointing *towards* one automatically emerges pointing *against*

Geozome

Recalling the aforementioned concept of the *Semantic Triple*, I can now apply it to the particulars within our geopolitical event. In such, situating, contextualizing and relating them. I can connect any entity to any other whilst having their relationship be defined by the triple's predicate. Continuing the connective process one is left with a construction which goes by many names. *Knowledge Graph*, *Semantic Network*, *Linked Data* are some of its aliases. I posit *Rhizome* may be another. In form and function, they are analogous.

Functionally, the "semiotic chains" constituting the Rhizome are parallels to our semantic triplet structure. They both denote relationships between things, people, qualities, quantities. They traverse at different "rates of flow," and bore through mediums of varying viscosities. The triplet structure thus provides us with the dictionary to the language spoken by the Rhizome. It realizes a "Rosetta Stone" of sorts that mediates conversation between the user, the computer and the geopolitical world.

Any Particulars the White House is connected to need to be uncovered in order to situate it in the Geopolitical Rhizome.

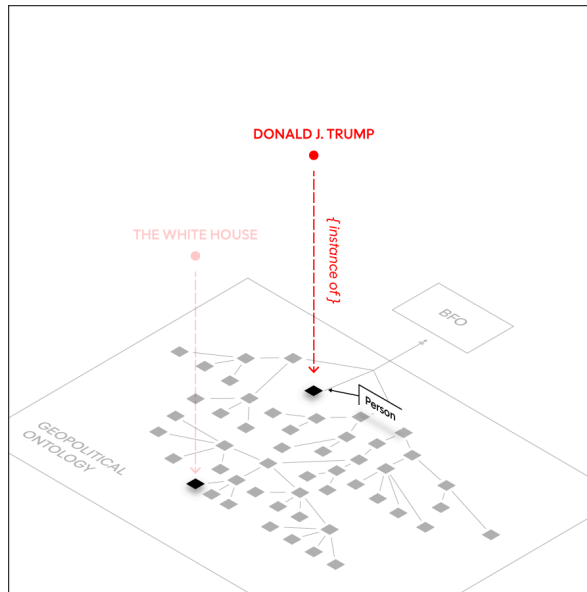


Fig. 15 - Donald Trump is an instance of the [Person] Universal.

Let us then connect the White House to its current tenant: Donald J. Trump, within our geopolitical rhizome. *Ontologically*, he is a [Person]. and is situated—through instantiation—with the following position.

The instantiation arrow's length is arbitrary as the predicate has no magnitude.

Each connective type (predicate) demands a clarification as to what these arrows can and cannot flow out from or flow into—their *Domain*¹⁶ and *Range*¹⁷.

¹⁶ **Domain** - The Universals a certain connection can originate from. The predicate "Can Eat" has the Domain of the Universal "Person."

¹⁷ **Range** - The Universals a certain connection can connect to. The predicate "Can Eat" has the Range of the Universal "Food."

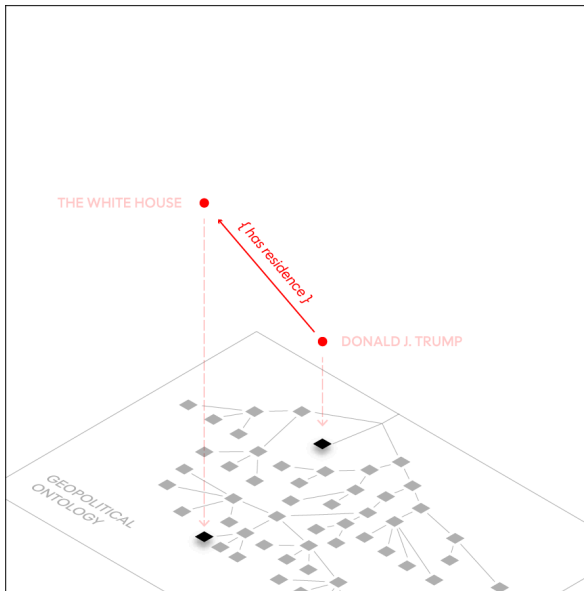


Fig. 16 - denoting that Donald Trump resides in the White House with the $\{has\ residence\}$ predicate.

I can now introduce a new predicate to denote their relationship: $\{has\ residence\}$. The predicate $\{has\ residence\}$ has the *domain* of the [Person] and the *range* of [Infrastructure]. A predicate's range and domain is inherited through the parent-child to any subclasses attached to it. Apropos, as [Infrastructure] is not a terminal class, all of its children classes can act as the range for the $\{has\ residence\}$ predicate.

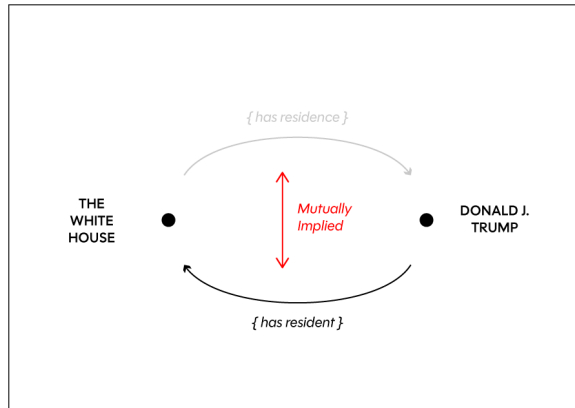


Fig. 17 - Biunivocal virality of rhizome and semantic triple.

The geopolitical rhizome demands to expand and multiply its connective tendrils. It is viral in that every arrow established linking one particular to another, explicitly implies a link directed in opposition to it. It forces the recognition of any relationship's dialectical implications. It allows for the user's geopolitical research to be documented from one perspective, and read from another. In an isolated case, it can elucidate very little. However, once I have formed long and entangled semiotic chains, the effect multiplies. I can seemingly move from one particular to another—following the novel semiotic routes—deemed disjointed.

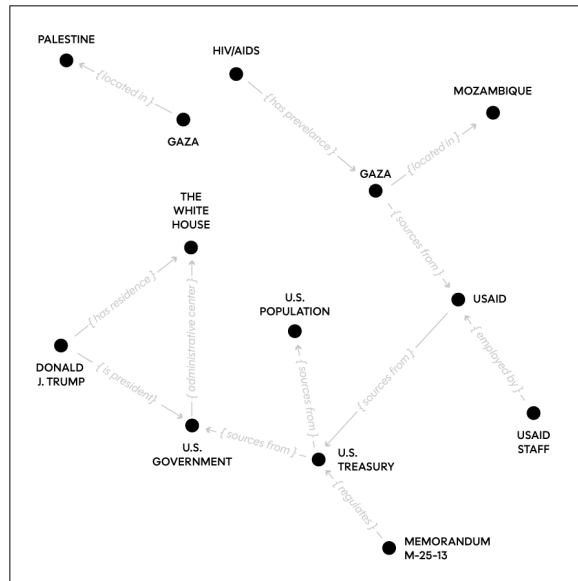


Fig. 18 - Miniature geopolitical rhizome (geozome)

I need to clarify that what I am constructing is a *map* of the geopolitical rhizome. Insofar as it is a limited representation of the world rather than the world itself. Allegorically, Jorge Luis Borges forwarns the diminishing returns of mapping at increasing precision in his tale about the king who built a 1:1 map of the kingdom. To over-exceedingly refine is to make it unnavigable. I am constructing a parochial approximation of *the Geopolitical Rhizome*. A "geozome".

I can populate our geozome with relevant particulars and connect them with our predicates from our ontology. Notice how once the ontology is out of sight, any intelligibility vanishes.

POSTULATE III

That the *geozome* is traversable.

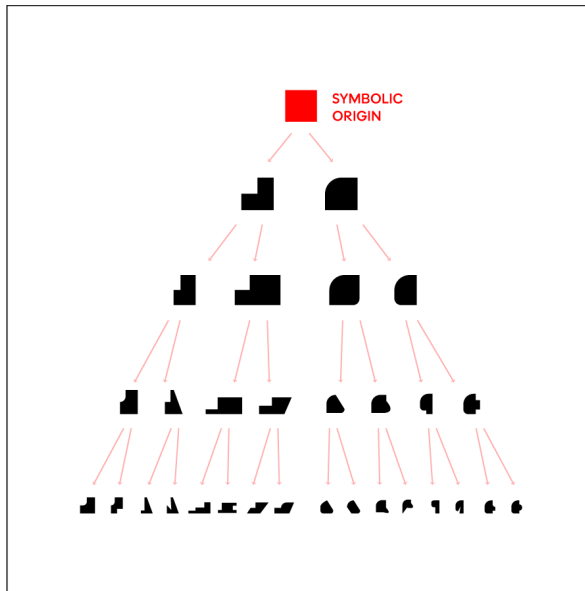


Fig. 19 - Example Symbol Ontology

Visualizing the *geozome*

Applying the *Aristotelian Logic*¹⁸ prescribed in the seminal guide for ontological-construction: *Building Ontologies with Basic Formal Ontology* (Arp et al. 69), I began the process of visualizing the *geozome*'s ontology. This was done through affixing the symbols used to represent the Universals to their location on the ontology. Meaning, I can select a Universal to serve as the *symbolic origin* and assign to it a graphical form. Every step descended from the *symbolic origin* visually manifests as a graphical modification to it's parent's symbol.

¹⁸ **Aristotelian Logic** - That "an [A] is a type of [B] that Cs"

i.e. A [Non-Consumable] is a [Processed Object] that *is consumed through use*.

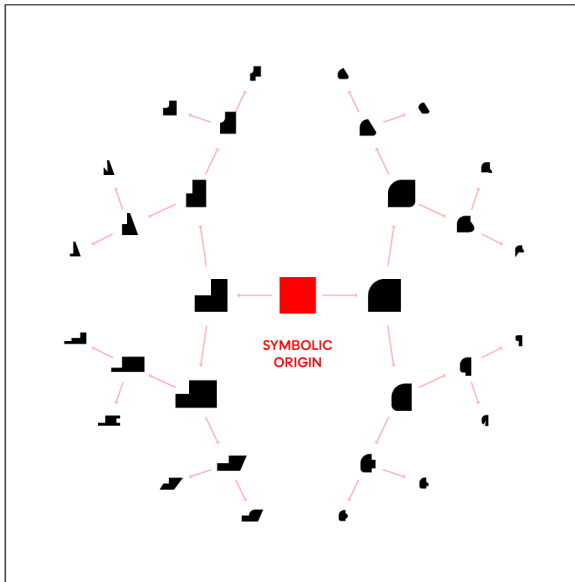


Fig. 20 - Radial presentation of ontology.

So far, I have only described the navigation of ontological structures as a *vertical* process. This is arbitrary and should not constrain our concept of how it should be read. We can just as easily imagine it as a construction of radial expansion. In so, framing *ontological specificity* as a function of distance from its origin, rather than extent downward.

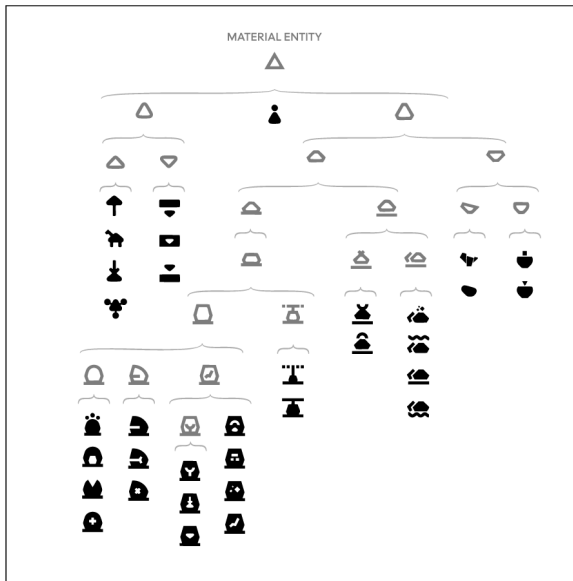


Fig. 21 - Geopolitical Symbol Ontology

Employing this method, I created a corresponding symbolic ontology that coats-in so, encoding visually-Universals and Particulars with their ontologic class. The following is an example of the fragment of the symbolic ontology which iconifies the [Material Entity] and it's children. The graphical form I selected as the symbolic origin was the triangle for its connotations to structural rigidity and integrity.

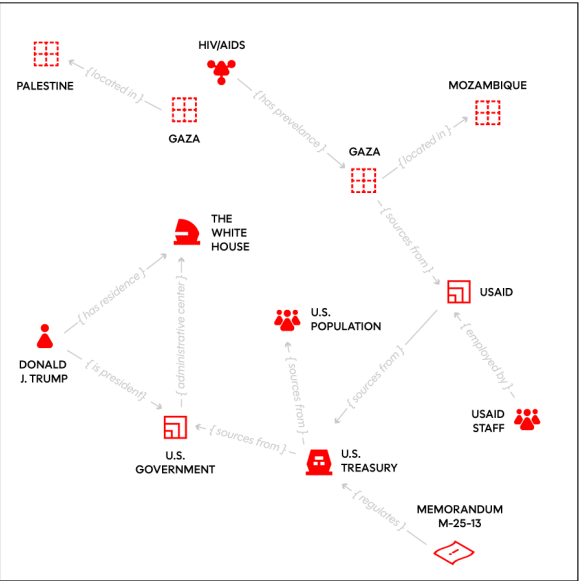


Fig. 22 - Symbolized geopolitical rhizome.

I can utilize our collection of symbols to make the geozome's content intelligible having detached it from the Universal's plane.

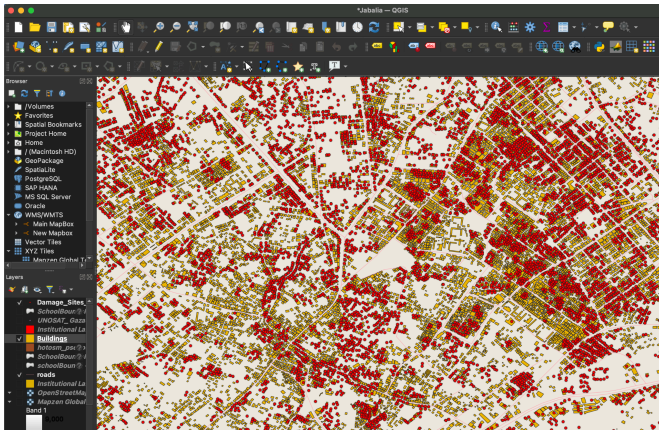


Fig 23 – Screenshot of QGIS displaying damaged sites in Jabalia, Gaza. (Depicted data was sourced from UNOSAT)

Traversing the *geozome*

Surveying the currently available digital platforms, illustrates the demand of a unified *Position* model. Each play to the niches of their user base. Geospatial Information Systems (GIS) software such as QGIS excel at analysis of data at a large scale. Patterns, hotspots and aberrations can all be discerned from the spatial frame it presents. It accounts for the spatial and temporal dimensions, but lacks a fleshed out and rigorous standard to account for the social dimension. It withholds guidelines for establishing what the imputed data *functionally is*, precisely because its absence is integral for GIS to be adopted for a wide breadth of applications. What GIS provides in scope, sacrifices in ontological clarity.

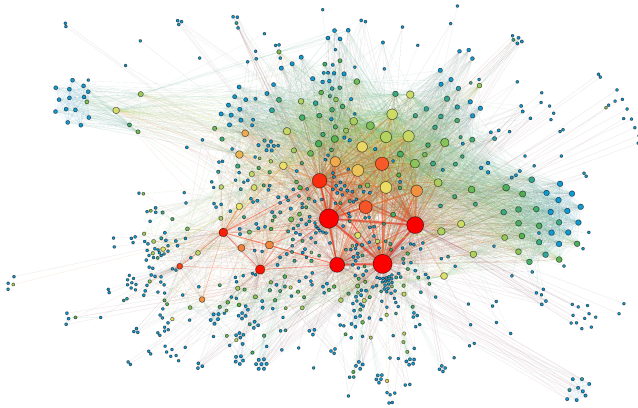


Fig 24 – Force-directed model of a social network. By Martin Grandjean, 13 April 2016

Conversely, we can examine applications that center social modeling and mapping. Network modeling applications like *Neo4j* or *Gephi* allow for a greater semantic grasp over the entities and their relations. Visualized through force-directed graphs, they can communicate social dynamics via attracting and repelling one another according to what their connection is. Redefining the concept of location by severing it from the geospatial. They employ a syntax similar to the *Semantic Triple* I mentioned earlier. However, without an accompanying ontology, any node is able to link to any other node, including illogical and inapplicable connections.

is a structure that requires *translation* (relocate) between its constituent entities, following available paths made by their connections. The translations within the geozome occur not in positions, but *Positions*.

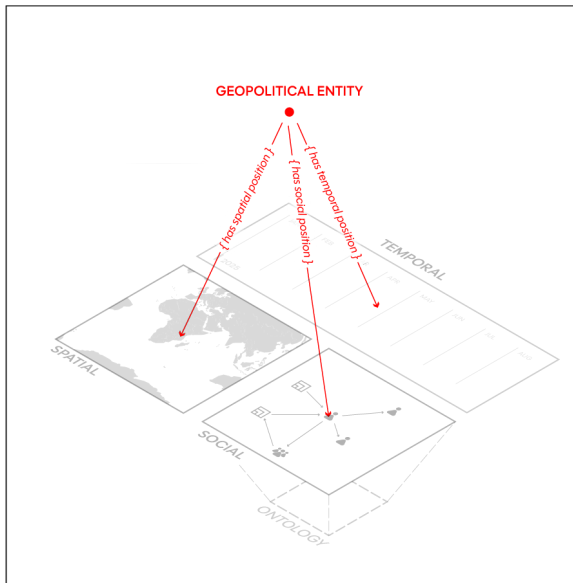


Figure X - Lorem Ipsum

The three components of the previously outlined *Position* now act as facets from which the *geozome* can be optically resolved from . Digital architecture fit to mediate this *translation* (reinterpret) between the user and the *geozome* is necessitated. In the context of the creation of a digital geopolitical mapping tool, these *Positional* dimensions manifest as separate platforms the *geozome* can be maneuvered around in. A movement in one of these dimensions influences the rest.

GeoZome is the interface I propose will facilitate the navigation of the *geozome*. It has 3 modes of moving through the *geozome*—one for every component of our *Positions*.

CONCLUSION

GeoZome

In summation, what mapping software needs foremost, is a foundational understanding of the information intended to be mapped. In the scope of geopolitics, it is inevitable that the information will have a spatial, a temporal and a social component. A map demands an allocation of its information to a position within some dimension. There exists a norm for the dimensionality of space (2D), and time (1D). In lieu of a norm for the social dimension, I posit that an ontological approach serves to establish one. By employing an ontology, we gain a unit of measurement (Universal), a position (Particular), and a method for traversal (Semantic Triple).

I discovered that the rhizome is a structure fit to model geopolitical events, agents and objects due to its properties and behavior. The rhizome (and thus, the geozome) is living,

consuming, ever-morphing and nauseatingly interconnected. It can capacitate both the material realities of resource extraction, and the immaterial realities of propaganda.

As I am modeling geopolitics with a rhizome, to take a “measurement” is to analyze its contextual links. Rhizomatic measurements seek noise rather than isolation. Geopolitics is composed of nothing but noise. As I am giving structure to the geozome with an ontology, I am afforded a syntax and vocabulary to vocalize and thus visualize the geozome. Semiotics is a form of navigation in that it incites traveling from your current context to another.

In GeoZome, this manifests through the Symbolic Ontology, wherein objects are not symbolized but what they look like but by their function. To *look* different is to *be* different. Geopolitically relevant in that it curtails the propensity to attribute be fooled by what an entity postures as rather than what function it serves.

If a police force harms civilians they definitionally transition from a corrective agent to a military one—even if their uniforms haven't changed.

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– Emilio Enrique Suárez Hode

A handwritten signature in black ink, reading "Emilio SH". The word "Emilio" is written in a cursive script, and "SH" is written in a more stylized, blocky cursive.